

human

Show me all the areas of trees within Exeter.

host_agent

Calling tool send_message with args : {'target': 'planning_agent', 'task_description': 'Show me all the areas of trees within Exeter.'}

planning_agent

The area is Exeter, so there is 1 entry for Exeter. The entity to search for is areas of trees, and I need to find as many entries as possible within Exeter.

output steps: ["Find Exeter 1 area", "Search for areas of trees in Exeter as many search results"]

Tool send_message returned result : The area is Exeter, so there is 1 entry for Exeter. The entity to search for is areas of trees, and I need to find as many entries as possible within Exeter.

output steps: ["Find Exeter 1 area", "Search for areas of trees in Exeter as many search results"]

Calling tool send_message with args : {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

named_area_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'}

Tool call_os_ngd returned result : ['Search results have been found. \n Artifact generated is : exeter_area_polygon,\n Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter.,\n count of records fetched : 98.\n ',]

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['exeter_area_polygon']}

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for polygons specifically representing Exeter
    # Use name1_text for 'Exeter', and also check for relevant descriptiongroup/description
    filtered = gdf[
        (
            (gdf['name1_text'].str.lower() == 'exeter') |
            (gdf['name1_text'].str.lower().str.contains('exeter'))
        ) & (
            (gdf['descriptiongroup'].isin(['Settlement', 'Part Of Settlement'])) |
            (gdf['description'].isin(['City', 'Town', 'Suburban Area', 'Village']))
        )
    ]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter using name and settlement/city/town/suburban/village criteria."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'GeoDataFrame containing only the polygon(s) specifically representing Exeter, filtered by name and relevant settlement/city/town/suburban/village attributes.'
    return [summary, artifact_name, artifact_description, filtered]
```

Tool code_executor returned result : ['Filtered to 1 polygon(s) for Exeter using name and settlement/city/town/suburban/village criteria.',]

Output from agent data_analysis_agent : Filtered to 1 polygon specifically representing Exeter using both the name and relevant settlement/city/town/suburban/village attributes. - Artifact name: filtered_exeter_area_polygon - Description: GeoDataFrame containing only the polygon(s) specifically representing Exeter, filtered by name and relevant settlement/city/town/suburban/village attributes.

Tool send_message returned result : ["Filtered to 1 polygon specifically representing Exeter using both the name and relevant settlement/city/town/suburban/village attributes.\n- Artifact name: filtered_exeter_area_polygon\n- Description: GeoDataFrame containing only the polygon(s) specifically representing Exeter, filtered by name and relevant settlement/city/town/suburban/village attributes.Additionally some data artifacts have been generated with names ['filtered_exeter_area_polygon'] and \n descriptions ['GeoDataFrame containing only the polygon(s) specifically representing Exeter, filtered by name and relevant settlement/city/town/suburban/village attributes.']", []]

Output from agent named_area : The specific polygon for Exeter has been filtered and saved as:

- **Artifact Name:** filtered_exeter_area_polygon
- **Description:** GeoDataFrame containing only the polygon(s) specifically representing Exeter, filtered by name and relevant settlement/city/town/suburban/village attributes.

Tool send_message returned result : ["The specific polygon for Exeter has been filtered and saved as:\n\n- **Artifact Name:** filtered_exeter_area_polygon\n- **Description:** GeoDataFrame containing only the polygon(s) specifically representing Exeter, filtered by name and relevant settlement/city/town/suburban/village attributes.Additionally some data artifacts have been generated with names ['filtered_exeter_area_polygon'] and \n descriptions ['GeoDataFrame containing only the polygon(s) specifically representing Exeter, filtered by name and relevant settlement/city/town/suburban/village attributes.']", []]

Calling tool send_message with args : {'target': 'land_features', 'task_description': "Search for all areas of trees within the artifact 'filtered_exeter_area_polygon'."}

land_features_agent

Calling tool call_os_ngd with args : {'filters': ['Coniferous Trees', 'Non-Coniferous Trees', 'Mixed Trees', 'Scattered Coniferous Trees', 'Scattered Non-Coniferous Trees', 'Scattered Mixed Trees'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'trees_in_exeter_area'}

Tool call_os_ngd returned result : ['Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are : ['land_trees_in_exeter_area'],\n Descriptions are : ["A geopandas dataframe containing land data with filters and bbox applied for the query Search for all areas of trees within the artifact 'filtered_exeter_area_polygon'. using filters ['Coniferous Trees', 'Non-Coniferous Trees', 'Mixed Trees', 'Scattered Coniferous Trees', 'Scattered Non-Coniferous Trees', 'Scattered Mixed Trees']. "],\n counts of records fetched are : [509].\n ', []]

Output from agent land_features : The search for areas of trees within the 'filtered_exeter_area_polygon' has been completed. Here are the results:

- **Artifact Name:** land_trees_in_exeter_area
- **Description:** A geopandas dataframe containing land data with filters applied for various types of trees within the specified area.
- **Number of Records:** 509

If you need further analysis or filtering, please let me know!

Tool send_message returned result : ['The search for areas of trees within the 'filtered_exeter_area_polygon' has been completed. Here are the results:\n\n- **Artifact Name:** land_trees_in_exeter_area\n- **Description:** A geopandas dataframe containing land data with filters applied for various types of trees within the specified area.\n- **Number of Records:** 509\n\nIf you need further analysis or filtering, please let me know!Additionally some data artifacts have been generated with names ['land_trees_in_exeter_area'] and \n descriptions ["A geopandas dataframe containing land data with filters and bbox applied for the query Search for all areas of trees within the artifact 'filtered_exeter_area_polygon'. using filters ['Coniferous Trees', 'Non-Coniferous

Trees\', \'Mixed Trees\', \'Scattered Coniferous Trees\', \'Scattered Non-Coniferous Trees\', \'Scattered Mixed Trees\']. "'], []

plotting_agent

Calling tool send_message with args : {'target': 'plotting_agent', 'task_description': "Plot all the areas of trees from the artifact 'land_trees_in_exeter_area' within the boundary of 'filtered_exeter_area_polygon'."}

Tool Code Executor

```
def plot_trees_within_exeter_boundary(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping

    # Unpack data
    trees_gdf = data[0].copy()
    exeter_gdf = data[1].copy()

    # Ensure both are in the same CRS (EPSG:4326 for folium)
    trees_gdf = trees_gdf.to_crs(epsg=4326)
    exeter_gdf = exeter_gdf.to_crs(epsg=4326)

    # Get the Exeter polygon (should be only one row)
    exeter_poly = exeter_gdf.iloc[0].geometry

    # Filter trees that are within the Exeter polygon
    trees_within = trees_gdf[trees_gdf.geometry.within(exeter_poly)]

    # Remove non-serializable columns (like Timestamps) for folium
    drop_cols = [col for col in trees_within.columns if 'date' in col or 'status_updatedate' in col]
    trees_within_simple = trees_within.drop(columns=drop_cols)

    # Create a folium map centered on Exeter
    centroid = exeter_poly.centroid
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)

    # Add Exeter boundary
    folium.GeoJson(mapping(exeter_poly), name='Exeter Boundary', style_function=lambda x: {'color': 'blue', 'weight': 2, 'fill': False}).add_to(m)

    # Add tree polygons
    folium.GeoJson(
        trees_within_simple,
        name='Tree Areas',
        style_function=lambda x: {'color': 'green', 'weight': 1, 'fillColor': 'green', 'fillOpacity': 0.4}
    ).add_to(m)
```

```
m.save('trees_within_exeter.html')

summary = f"Plotted {len(trees_within)} tree areas within the Exeter boundary. First 5 osid: {trees_within['osid'].head(5).tolist()}"
artifact_name = 'trees_within_exeter.html'
artifact_description = 'A folium map showing all tree polygons from land_trees_in_exeter_area that are within the filtered_exeter_area_polygon boundary.'
artifact_data = 'trees_within_exeter.html'
return [summary, artifact_name, artifact_description, artifact_data]
```

Tool code_executor returned result : ["Plotted 490 tree areas within the Exeter boundary. First 5 osid: ['50194858-41d8-4dca-acc9-67d7328cd145', '25248ceb-9d3b-40ad-ada8-213048d2cc50', 'd407f2c1-07df-4e45-afea-97b3e135eb51', 'a387423b-272c-4038-8cd9-3b7b5d04fa0e', 'bd31b9ba-205c-4ae0-922b-ca854aab2b60']",]

Output from agent plotting_agent : Plotted 490 tree areas within the Exeter boundary. First 5 osid: ['50194858-41d8-4dca-acc9-67d7328cd145', '25248ceb-9d3b-40ad-ada8-213048d2cc50', 'd407f2c1-07df-4e45-afea-97b3e135eb51', 'a387423b-272c-4038-8cd9-3b7b5d04fa0e', 'bd31b9ba-205c-4ae0-922b-ca854aab2b60']

Artifact name: trees_within_exeter.html

Artifact description: A folium map showing all tree polygons from land_trees_in_exeter_area that are within the filtered_exeter_area_polygon boundary.

Artifact data: trees_within_exeter.html

host_agent

All the areas of trees within Exeter have been identified and mapped. There are 490 distinct tree areas, including coniferous, non-coniferous, mixed, and scattered tree types, all within the official Exeter boundary.

- The map artifact is named: trees_within_exeter.html
- Description: A folium map showing all tree polygons from the land features dataset that are within the Exeter area.

If you would like to view or analyze this map further, please let me know!